Automated Orthopedic Functional Assessment

AI-Based 3D Human Pose Estimation for Automated Clinical ROM Analysis

Presenter: **Sara Amiri**

Supervisor: **Dr. Anna Vögele**

Functional Assessment in Orthopedic Rehabilitation (The Problem)

The Clinical Problem Domain:

Post-surgical rehabilitation monitoring:

- **Knee replacement**
- **Hip reconstruction**
- **Spinal surgery**

Functional Assessment in Orthopedic Rehabilitation (The Problem)

Current Manual Workflow

System

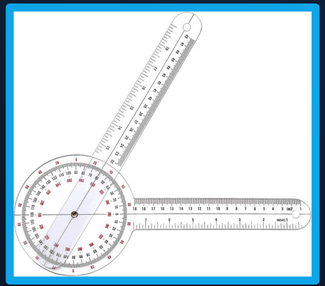
- Mechanical Goniometer + Human Eye

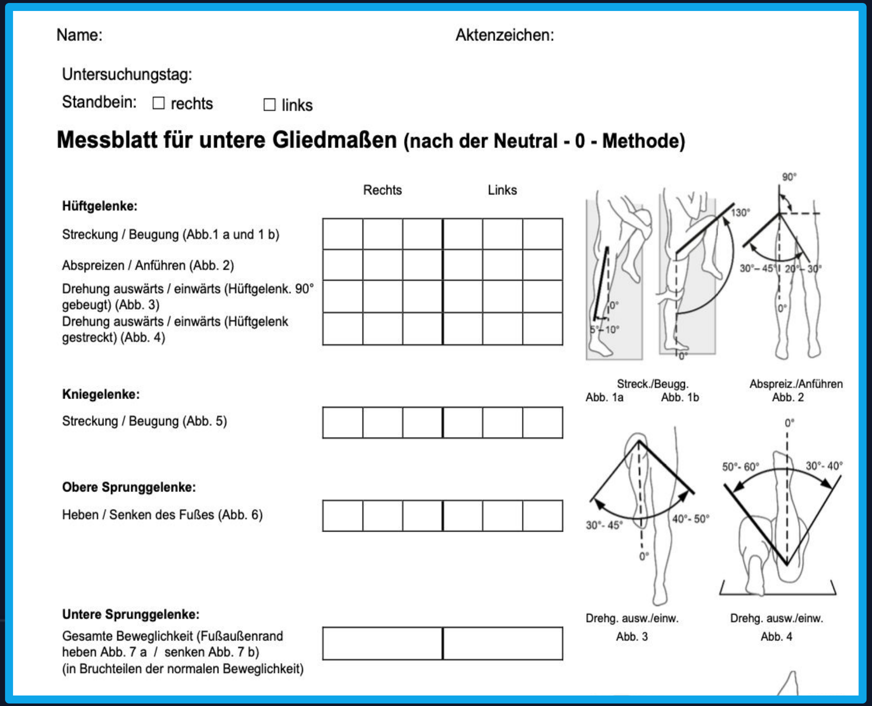
Workflow

- Measure joint angle → Record Neutral-0 values → Fill paper documentation

Limitations:

- manual and time-consuming
- subjective measurements
- requires in-person visits





Name: Aktenzeichen:

Untersuchungstag:

Standbein: ☐ rechts ☐ links

Messblatt für untere Gliedmaßen (nach der Neutral - 0 - Methode)

	Rechts			Links		
Hüftgelenke:						
Streckung / Beugung (Abb.1 a und 1 b)						
Abspreizen / Anführen (Abb. 2)						
Drehung auswärts / einwärts (Hüftgelenk. 90° gebeugt) (Abb. 3)						
Drehung auswärts / einwärts (Hüftgelenk gestreckt) (Abb. 4)						
Kniegelenke:						
Streckung / Beugung (Abb. 5)						
Obere Sprunggelenke:						
Heben / Senken des Fußes (Abb. 6)						
Untere Sprunggelenke:						
Gesamte Beweglichkeit (Fußaußenrand heben Abb. 7 a / senken Abb. 7 b) (in Bruchteilen der normalen Beweglichkeit)						

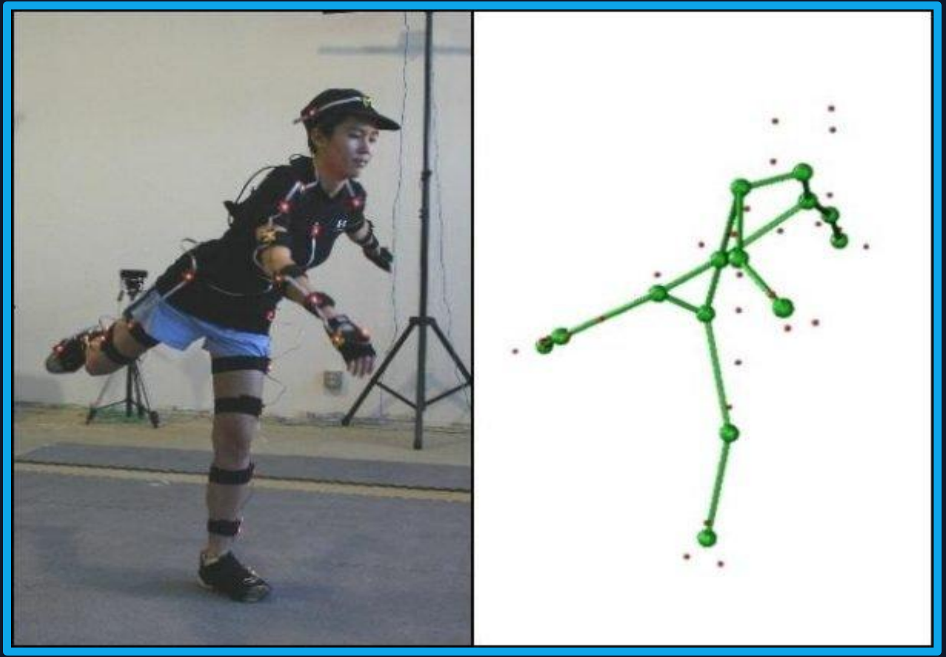
Streck./Beugg. Abb. 1a Abb. 1b

Abspreiz./Anführen Abb. 2

Drehg. ausw./einw. Abb. 3

Drehg. ausw./einw. Abb. 4

Functional Assessment in Orthopedic Rehabilitation (The Problem)



Related Marker-Based Workflow

System

- Marker-based optical motion capture (Vicon / OptiTrack)

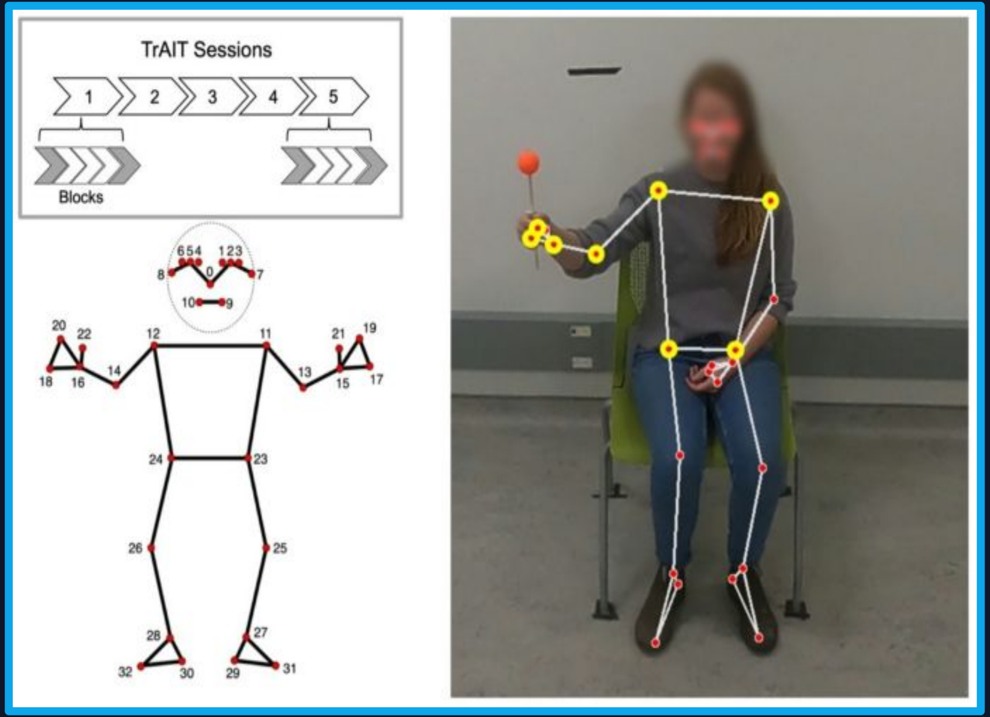
Pros:

- Highly accurate
- Gold-standard motion tracking

Cons:

- Expensive
- Specialized lab required
- Long setup time

Functional Assessment in Orthopedic Rehabilitation (The Problem)



Related Marker-Less Workflow

System

- **Markerless baseline tracking (Google MediaPipe)**

Pros:

- **Zero-cost / Open-source**
- **Runs in real-time on standard consumer hardware (CPU-friendly)**
- **Instant setup with no calibration or skin markers required**

Cons:

- **No absolute real-world millimeter coordinates (X, Y, Z)**
- **Not able to track complex spinal bending**

AI-Based Automated ROM Assessment (The Solution)

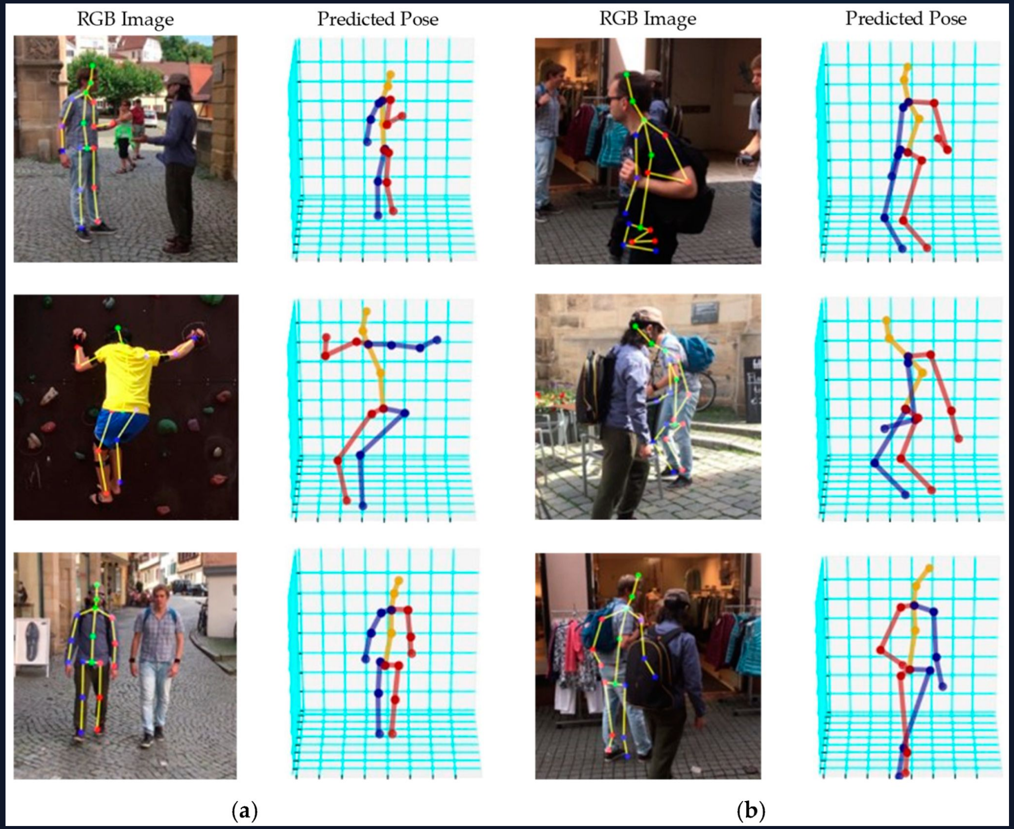
Proposed Automated Solution:

Markerless 3D motion capture using a standard webcam:

- automatic human pose estimation
- automatic joint angle calculation
- digitalization of orthopedic assessment workflows

Technical Approach

- video captured from a normal webcam
- 3D pose estimation using MeTRAbs
- extraction of body landmarks and joint positions
- calculation of orthopedic angles and Neutral-0 measurements



AI-Based Automated ROM Assessment (The Solution)

Advantages

- low-cost
- objective and reproducible measurements
- reduced manual workload
- supports remote rehabilitation and telemedicine

Application Area:

- Computer Vision
- Digital Health
- Clinical Automation
- Telemedicine

Contribution & Motivation

The Problem Solved

- Eliminates manual subjectivity and reading errors.
- Breaks the location dependency

Core Beneficiaries

- Clinicians: Automated digital charting.
- Patients: Lab-grade Telemedicine from home via consumer webcams.

The Contribution

A custom 3D Geometric Analysis Pipeline mapping spatial coordinates (X,Y,Z) to clinical Neutral-0 triples.

Technical Novelty

- Achieving Markerless Lab Accuracy via monocular video.
- Supports complex articulated spinal models where competitors fail.

Core Component: MeTRAbs Integration

Fine-Tuning MeTR[illegible]s model for Automated Clinical ROM Analysis:

- Integration of the MeTRAbs 3D Human Pose Estimation framework.

- Input: RGB Video
- Output: 3D Joint Coordinates (X, Y, Z) in millimeters

- Technique: Using an Affine-Combining Autoencoder (ACAE) to model a complex, articulated human spine topology.
    - **ACAE:** a neural-network-based structure used to learn complicated body movement relationships automatically.

Data Sources & Resources

Pre-training Pool

MeTRAbs utilized large-scale datasets for diverse pose supervision:

- COCO / SURREAL
- MuCo-3DHP
- CMU Panoptic / AIST++

Validation Baseline

Human3.6M

provides highly accurate 3D ground-truth annotations:

- Millimeter-Scale Precision
- Highly Controlled Environment
- Scientific Validity

Clinical Data

Reimer et al. (2021)

Anonymized orthopedic recordings covering functional tools and spinal mobility assessments.

Processing & Generated Records

Monocular RGB Input → MeTRAbs 3D Estimation → Coordinate Extraction (X,Y,Z) → Geometric Angle → Neutral-0 Metrics

Generated Records

- 3D joint trajectories
- Anatomical angles
- Neutral-0 metrics
- Digital rehab logs

Access & Permissions

- H3.6M: Academic License via Uni Bonn affiliation.
- Clinical Data: Institutional agreement; Research use only.

Methods & Technical Infrastructure

Intended Methods

- **3D Pose Lifting:** Absolute metric-scale joint reconstruction.
- **Vector Geometry:** Custom anatomical angle mapping.
- **Temporal Filtering:** Butterworth/Savitzky-Golay for jitter reduction.

Technical Stack

- **Language:** Python
- **Frameworks:** TensorFlow (MeTRAbs), OpenCV.
- **Libraries:** NumPy, SciPy, Matplotlib.

APIs & Toolkits

- **Articulated Model:** ACAE (Affine-Combining Autoencoder).
- **Neutral-0 Engine:** Custom medical-to-geom[illegible] API.

Methods & Technical Infrastructure

Category	Requirement / Resource	Access & Licensing
Hardware / GPU	NVIDIA GPU (RTX 30-series or higher) with CUDA 11.x / cuDNN support.	Local High-Performance Workstation provided by the Group.
Software Licenses	MIT (MeTRAbs), Apache 2.0 (TF), GPL (OpenCV).	All frameworks are Open Source and research-compliant.
Dependencies	Standard Conda/Pip environment management.	Version-locked requirements for clinical reproducibility.

Primary Evaluation Goal: Clinical Equivalence

To demonstrate that a monocular, markerless AI pipeline can achieve medical-grade measurements equivalent to laboratory MoCap within a **< 5.0° MAE margin**.

Planned Tests & Evaluation

Accuracy Evaluation

Comparison against Human3.6M ground truth:

- Mean Absolute Error (MAE)
- validation of 3D joint reconstruction
- validation of Neutral-0 angle estimation

Performance Benchmarks

Evaluate computational efficiency:

- inference latency
- frames per second (FPS)
- GPU vs. CPU execution

Robustness Testing

Test system stability under challenging conditions:

- loose clinical clothing
- low-light environments
- partial joint occlusion
- non-ideal camera positioning

Reproducibility Checks

Repeated execution of identical recordings to verify:

- stable 3D coordinate extraction
- consistent anatomical angle measurements

Usability Evaluation

Clinical workflow assessment:

- time required to obtain a valid Neutral-0 measurement
- comparison with manual goniometry workflow
- ease of use in rehabilitation settings

Portability & Stress Testing

cross-platform execution (Linux / Windows)

Conda environment reproducibility

extended recording sessions

real-time tracking stability